Number Sequences & Patterns

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. 15, 18

2. 50, 60

3. B — 10

4. 10

5. 35, 30

6. 9, 11

7. A, C and E — 5,10,15,20 / 15,20,25,30 / 30,35,40,45 all step up by 5

8. 12

9. 70, 60

10. B — add 6

11. 14, 17

12. 20

13. A, C and E — 45, 54 and 63 are multiples of 9

14. 28, 35

15. C — 24

Standard

16. 32, 64

17. 16, 32

18. C — double each time

19. 10

20. 10, 5

21. 7 squares

22. A, B and D — each steps up by 4

23. 15

24. 65, 55

25. B — 22

26. 55, 66

27. 48

28. 70

29. A — add consecutive odd numbers

30. A, C and E — 9, 16 and 25 are square numbers

Standard / Hard

31. 24, 29 (add 5 each time)

32. C — 25 (add 5 each time)

33. 4 + 2×4 = 12 people

34. 7 (add 4 each time)

35. A, C and E (each has a constant difference; B doubles each time and D's differences increase)

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36. 3 + 2×3 = 9 sticks

37. 25, 36 (square numbers)

38. C — 25 (not a multiple of 6)

39. 40, 35, 30, 25 — the 4th term is 25

40. 16 (add 6 each time)

41. 4 + 3×4 = 16 edges

42. A and C

43. 36 (subtract 6 each time)

44. C — 48 (doubling)

45. 7 ((4+5)−2)

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Homework Worksheet — Answers

Easy / Standard

1. 20, 24

2. 80, 100

3. C — 25

4. 12

5. 60, 50

6. 13, 16

7. A, C and E

8. 24

9. 42, 36

10. B — add 7

11. 23, 28

12. 30

13. B and D — 40 and 48

14. 36, 45

15. C — 26

Standard

16. 48, 96

17. 81, 243

18. C — double each time

19. 11

20. 20, 10

21. 18 sides

22. A, C and E — each steps up by 6

23. 17

24. 55, 45

25. B — 18

26. 60, 72

27. 162

28. 64

29. A — add consecutive odd numbers

30. A, C and E — 16, 36 and 49 are square numbers

Standard / Advanced

31. 4, 7, 10, 13, 16 — the 5th term is 16

32. 6 (3→4→6: double then subtract 2 each step)

33. B — 17 (2, 7, 12, 17 — add 5 each time)

34. A, C and E (the sequence is all odd numbers: 1,3,5,7,9,11,13,15,17,19,21…; 8 and 20 are even, so not terms)

35. 10 + 3×4 = 22 seats

36. The 8th term, −6 (50 − 8×7 = −6)

37. 4 + 7 + 10 = 21 tiles

38. A — Yes, because 3 + 4×9 = 39

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39. 7 weeks (5 + 4×7 = 33, which is more than £30; after 6 weeks it is only £29)

40. A: 3, 8, 13, 18, 23. B: 11, 14, 17, 20, 23. Yes — they are both 23 as their 5th listed term

41. 4 + 3×5 = 19 sticks

42. A — the 10th term is 6 + 5×9 = 51

43. The 6th triangular number

44. A, B, C and D (common difference is 4; 6th term = 7 + 4×5 = 27, so E is wrong; both 51 and 31 fit the pattern 7 + 4×(term number − 1) for a whole term number)

45. 14 ((4+3)×2 = 14)

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